IDEAGOV - International Center for Decentralization and Governance

Working Paper 2026-14

The Impact of Population Ageing on Public Finances in the EU-27

María Cadaval-Sampedro
Santiago Lago-Peñas
Xoaquín Fernández-Leiceaga
Alejandro Domínguez-Lamela

July, 2026

IDEAGOV – International Center for Decentralization and Governance
Facultade de Ciencias Económicas e Empresariais (USC)
Av. Do Burgo das Nacións, s/n. Campus Norte. 15786 Santiago de Compostela. Spain.
ideagov@ideagov.eu
www.ideagov.eu

IDEAGOV is an international hub for research and policy dissemination on innovations in decentralization and governance, and a leading reference for the study of federalism within the European Union and globally, combining the backgrounds of economists, political scientists and geographers.

IDEAGOV – International Center for Decentralization and Governance
Facultade de Ciencias Económicas e Empresariais (USC)
Av. Do Burgo das Nacións, s/n. Campus Norte. 15786 Santiago de Compostela
Spain
ideagov@ideagov.eu
www.ideagov.eu

The Impact of Population Ageing on Public Finances in the EU-27

María Cadaval-Sampedro[1,2,3], Santiago Lago-Peñas[1,2,3], Xoaquín Fernández-Leiceaga[1,2,3], and Alejandro Domínguez-Lamela[1,2]

[1] Universidade de Santiago de Compostela (USC)
[2] International Center for Decentralization and Governance (IDEAGOV)
[3] CISPAC: Research on Territory

ABSTRACT

The paper analyses the impact of demographic ageing on age-related public expenditure in the EU-27 over the period 1992-2024. The analysis covers healthcare, long-term care (LTC), and pensions. The results reveal three patterns. First, demographic effects are heterogeneous across countries, driven by short-run responses that vary by spending category—limited in healthcare, absent in LTC. Second, advanced ageing (80+) tends to be more important for long-run dynamics, especially in healthcare. Third, Central and Eastern European economies show weaker or delayed responses, while Western and Nordic countries exhibit stronger adjustments, more evident in healthcare and LTC than in pensions, reflecting the decisive role of institutional design.

Keywords: Ageing, public expenditure, demographic change, European Union

JEL classification: E60, E62, H20, H51, H55, J11, J14

This paper benefitted from funding project 20250107, "The impact of poputalion ageing on public accounts in the EU-27", of the Spanish Institute of Fiscal Studies – IEF within the research line "Budget and Public Expenditure".

1. INTRODUCTION

Population ageing is one of the principal drivers of public finance dynamics in advanced economies. The combination of persistently low fertility rates, sustained gains in life expectancy, and weak demographic growth, insufficiently offset by migration, has led to a steady increase in the share of individuals aged 65 and over in the working-age population. This demographic transformation has become a central concern for policymakers and researchers alike, given its potential implications for the sustainability of public finances and welfare systems.

The fiscal consequences of ageing operate through multiple channels. On the expenditure side, it increases demand for pensions, healthcare services, and long-term care. On the revenue side, demographic change tends to reduce labor-force participation and weaken labor-based tax bases, thereby affecting governments' tax revenues. The issue is particularly relevant in the EU, where member states face broadly similar demographic trends. However, countries differ in their institutional arrangements, welfare-state models, and public policy frameworks, which may condition how demographic pressures are translated into public expenditure. Consequently, similar demographic developments may generate markedly different outcomes.

Understanding how demographic change translates into public expenditure pressures across different institutional settings is essential for assessing the long-term implications of ageing and designing effective policy responses. A substantial body of literature has examined this relationship, generally finding that ageing increases pressure on public expenditure, particularly through pensions, healthcare, and long-term care. However, most studies focus on individual expenditure categories, specific countries, or limited groups of countries, making it difficult to determine whether the fiscal consequences of ageing follow common patterns across Europe or are largely shaped by national institutional characteristics.

Therefore, the main contribution of this paper to the existing literature is twofold. First, it adopts a cross-country comparative perspective, examining the impact of population ageing on age-related public expenditure in the EU-27 countries over the period 1992-2024. Second, the paper explicitly addresses cross-country heterogeneity through a combination of dynamic Error Correction Models (ECM), mixed-effects estimations, and cluster analysis. In doing so, it provides new evidence on the extent to which institutional differences shape the expenditure consequences of demographic ageing across European countries.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature. Section 3 presents the data and variables. Section 4 describes the econometric methodology. Section 5 discusses the empirical results. Finally, section 6 concludes and outlines the main policy implications.

2. LITERATURE REVIEW

Population ageing has become a central structural determinant of fiscal dynamics in advanced economies. A broad empirical consensus identifies demographic change as a key driver of long-term fiscal imbalances, although the magnitude and transmission mechanisms of these effects remain highly heterogeneous across countries and methodological approaches (OECD, 2022a; European Commission, 2024; IMF, 2023).

The theoretical foundations of this literature are primarily grounded in life-cycle theory and overlapping-generations (OLG) models (Auerbach et al., 1994; Auerbach and Kotlikoff, 1999), which formalise the intertemporal distribution of fiscal contributions and transfers across cohorts. In these frameworks, individuals are net contributors during working ages and net beneficiaries in retirement, implying that increases in old-age dependency ratios generate a structural deterioration in fiscal balances. OLG models also emphasise broader macroeconomic channels, including changes in savings behaviour, labour supply, productivity, and public debt dynamics under intertemporal budget constraints (Beetsma et al., 2025; Ramos-Herrera and Sosvilla-Rivero, 2020).

The empirical literature identifies two primary transmission channels. On the revenue side, population ageing erodes the labour tax base by reducing the working-age population, lowering labour-force participation, and moderating wage growth. Given the centrality of labour taxation in OECD and EU fiscal systems, this mechanism leads to a structural weakening of revenue capacity. A substantial body of panel-data evidence documents a negative association between ageing indicators and fiscal revenues or overall budget balances (Filipović & Miljković, 2024; OECD, 2022a), although estimated effects vary across institutional contexts. In contrast, consumption and capital taxation appear less directly affected, reflecting differences in behavioural responses, tax design, and the relatively low taxation of wealth (Benzarti and Carloni, 2019; Piketty and Saez, 2013; OECD, 2021).

On the expenditure side, ageing is consistently associated with rising public spending, particularly in pensions, healthcare, and long-term care. Pension expenditure exhibits the most direct and robust relationship with demographic structure, especially in pay-as-you-go systems where fiscal sustainability depends critically on the contributor-beneficiary ratio (Verbič & Spruk, 2024). Empirical evidence confirms a positive elasticity between ageing indicators and pension spending, although parametric reforms, such as increases in statutory retirement ages and adjustments to benefit formulas, have partially mitigated these pressures (Aguilar-Segado, 2024; Szwarc, 2024). Healthcare and long-term care expenditures are also expected to increase, driven by higher prevalence of chronic conditions, end-of-life costs, and growing demand for care services, although their dynamics are shaped by non-demographic factors such as technological progress, institutional design, and efficiency (Cylus et al., 2022; OECD, 2024; Navarro-García and Sarria-Santamera, 2023).

Beyond these direct fiscal effects, a growing strand of literature highlights indirect macroeconomic channels through which ageing affects fiscal sustainability. Demographic change is associated with lower potential output due to reductions in labour supply, declining productivity growth, and shifts in sectoral composition towards less productive activities (Aksoy et al., 2019; Beetsma et al., 2025). These dynamics weaken fiscal capacity by simultaneously reducing tax bases and economic growth. However, the net effect remains ambiguous, as labour scarcity may induce capital deepening, automation, and technological adoption, partially offsetting productivity losses (Acemoglu and Restrepo, 2022).

A central insight of the empirical literature is the substantial heterogeneity in the fiscal effects of ageing. Cross-country differences in pension systems, labour-market institutions, tax structures, and healthcare organisation significantly mediate the impact of demographic change (Brys et al., 2016; Petrucci and Margarito, 2025). As a result, similar demographic trends can produce markedly different fiscal outcomes depending on institutional configurations. This heterogeneity is reflected in empirical findings, which range from strongly negative effects to statistically insignificant relationships once macroeconomic controls and institutional variables are included (Twarowska and Twarowska-Mól, 2024).

Methodological challenges further complicate empirical identification. Many studies rely on aggregate demographic indicators that fail to capture heterogeneity in health status, effective labour supply, or cohort composition. Moreover, endogeneity concerns are pervasive, as fiscal variables and demographic dynamics are jointly determined through feedback mechanisms, including those linking public expenditure, health outcomes, and longevity (Ajovalasit et al., 2024; Iuga et al., 2024). While recent contributions employ dynamic panel models, system-GMM estimators, and stochastic projections, causal identification remains a central challenge.

Recent research has also incorporated additional moderating factors. Migration and increased labour-force participation may partially offset demographic pressures, although their long-term impact appears limited (OECD, 2020; Michalski and Stępień, 2021). Improvements in population health may delay retirement and reduce expenditure pressures (Iuga et al., 2024), while political economy constraints -particularly the increasing electoral weight of older cohorts- may hinder the implementation of fiscal reforms (Brosig, 2022; Barilari et al., 2024).

3. EMPIRICAL STRATEGY

Our empirical work is based on an unbalanced panel dataset covering 27 European Union countries over the period 1992–2024. The main data source is Eurostat, ensuring consistency in definitions and comparability across countries and over time. The analysis focuses on the three main public expenditure categories related to ageing: healthcare (h), long-term care (ltc), and pensions (p), as well as their aggregate (a = h + ltc + p). All of them are expressed as a percentage of GDP.

The key regressors capture demographic patterns. We combine two variables: while the share of the population aged 65 and over reflects the extensive dimension of ageing, the share of the population aged 80 and over, relative to the elderly population, captures the intensive one. The econometric specification also includes a set of standard macroeconomic controls: real GDP growth, GDP per capita in purchasing power standards, total government expenditure as a share of GDP, and the fiscal balance (deficit/surplus). These variables account for cyclical and structural factors affecting public spending dynamics. The last three are lagged by one period to avoid simultaneity bias. A dummy variable is also introduced to capture the COVID-19 period (2020–2021), which is expected to have had a substantial impact on public expenditure over GDP. First, because of the drop in GDP in most countries. Second, the required additional healthcare spending. A detailed description of all variables and their definitions is provided in **Table 1**.

Table 1. Variable definitions

Variable	Description	Measurement	Source	Period
a	Total public age-related expenditure [a + ltc + p]	% of GDP	Eurostat	1992-2023
h	Public expenditure on Healthcare	% of GDP	Eurostat	1992-2024
ltc	Public expenditure on Long-Term Care	% of GDP	Eurostat	1992-2024
p	Public expenditure on Pensions	% of GDP	Eurostat	1992-2023
pc65	Share of population aged 65 and over	% of total population	Eurostat	1992-2024
p80	Share of population aged 80 and over relative to population aged 65+	% of aged population aged 65 and over	Eurostat	1992-2024
growth	Real GDP growth rate	percentage point change	Eurostat	1992-2024
gdp	GDP per capita in PPS	EU27 = 100	Eurostat	1995-2024
exp_gdp	General government expenditure	% of GDP	Eurostat	1992-2024
def	Net lending / net borrowing	% of GDP	Eurostat	1995-2024

To capture both the persistence of public expenditure and the effects of demographic ageing, public expenditure is modelled as a function of its own lagged value and the variables listed above. We combine Error Correction Models (ECM) for Time-Series Cross-Section (TSCS) data and mixed-effects specifications to account for unobserved heterogeneity, dynamic adjustment, and cross-country differences in responses.

We model the dynamics using an error-correction specification, even though the estimated autoregressive coefficient on the lagged dependent variable is around 0.8. This choice follows the general argument in the TSCS literature that error-correction formulations are a convenient reparameterization of autoregressive distributed lag models, allowing a transparent separation between short-run responses and long-run equilibrium adjustment, even when the process is stationary but highly persistent. Moreover, to assess the time-series properties of the data, we performed standard unit root tests on the dependent variable. In most cases, the null of a unit root is rejected at conventional significance levels, indicating stationarity, while in one specification the test fails to reject the unit root only at the 10 per cent level, which we interpret as being consistent with a highly persistent, near-integrated process. In this context, the error-correction parameterization provides an interpretable summary of the speed of adjustment and long-run effects without relying mechanically on binary unit-root classifications. This is particularly useful in TSCS settings, where the substantive interest often lies in both short-run responses and gradual adjustment over time (Beck and Katz, 2011).

The ECM is estimated using Least Squares Dummy Variables (LSDV) with country-specific effects, insofar as it offers a parsimonious and interpretable way to model persistence, long-run adjustment, unit-specific intercepts, and heterogeneous effects.[1]

The ECM specification is given by:

$$\Delta \mathrm{exp}_{it} = \alpha\, \mathrm{exp}_{it-1} + \beta_1 \mathrm{pc65}_{it-1} + \beta_2 \mathrm{p80}_{it-1} + \lambda_1 \Delta \mathrm{pc65}_{it} +$$

$$+\lambda_2 \Delta \mathrm{p80}_{it} + \delta' \Delta \mathrm{X}_{it} + +\delta \mathrm{COVID}_t + \mu_i + \varepsilon_{it} \qquad [1]$$

where:

- $\exp_{it}$ denotes public expenditure (healthcare, long-term care, pensions, or total expenditure) expressed as a percentage of GDP
- $pc65_{it}$ is the share of the population aged 65 and over
- $p80_{it}$ is the share of the population aged 80 and over relative to the elderly population
- X_{it} is the vector of macroeconomic controls
- $COVID_t$ is a dummy variable equal to 1 for the years 2020–2021[2]
- μ_i captures unobserved country-specific effects
- ε_{it} is the idiosyncratic error term

The inclusion of the lagged dependent variable captures the strong persistence typically observed in public expenditure. Fixed effects control for time-invariant country characteristics such as institutional design or welfare state structure.

- β parameters capture long-run effects through differenced variables (Δ)
- λ parameters capture short-run effects through lagged levels $(it - 1)$

However, a homogeneity test of slopes clearly rejects the null hypothesis of common coefficients across countries for all expenditure variables. The homogeneous specification would therefore be interpreted, at best, as an average effect across countries, potentially masking important differences in the magnitude or even the direction of the relationship. For this reason, the next step is to rely on econometric approaches that allow the estimated coefficients to vary across countries. A mixed-effects (random-coefficients)

---

[1] The inclusion of a lagged dependent variable in a fixed-effects model generates the well-known Nickell bias. Still, this bias is of order (1/T) and is therefore primarily troublesome in very short panels, when T is 3 or 4 (Nickell 1981). Since the average time dimension in our data exceeds 15, this concern is mitigated. Moreover, Beck and Katz's Monte Carlo evidence shows that, for the values of (T) typically encountered in TSCS applications, roughly 20 or more, OLS/LSDV with fixed effects performs about as well as the Kiviet correction and substantially better than the Anderson-Hsiao estimator (Beck and Katz 2011). This supports the use of LSDV in our setting, especially because it remains flexible enough to accommodate country-specific intercepts and slope heterogeneity. With only 27 countries, dynamic panel GMM is not an attractive baseline estimator. Arellano-Bond and related GMM estimators are designed for large-N, small-T panels. In our setting, the advantages of GMM are limited, while the risks of instrument proliferation, weak specification tests, and loss of flexibility are substantial. We therefore prefer an LSDV-based dynamic specification.

[2] Alternative specifications with period fixed effects were also estimated, yielding qualitatively similar results. We choose the COVID-19 dummy solution as a more parsimonious way to capture the exceptional fiscal dynamics during the pandemic period.

model is estimated using restricted maximum likelihood (REML), which provides consistent estimates of variance components (Baltagi, 2021). This approach enables the recovery of country-specific coefficients that can subsequently be used in a clustering analysis.[3]

The specification allows the coefficients associated with demographic variables to vary across countries[4]:

$$\Delta \mathrm{exp}_{\mathrm{it}} = \alpha\ \mathrm{exp}_{\mathrm{it-1}} + \beta_{1\mathrm{i}} \mathrm{pc65}_{\mathrm{it-1}} + \beta_{2\mathrm{i}} \mathrm{p80}_{\mathrm{it-1}} + \lambda_{1\mathrm{i}} \Delta \mathrm{pc65}_{\mathrm{it}} + \\ + \lambda_{2\mathrm{i}} \Delta \mathrm{p80}_{\mathrm{it}} + \delta' \Delta \mathrm{X}_{\mathrm{it}} + \delta \mathrm{COVID}_{\mathrm{t}} + \mu_{\mathrm{i}} + \varepsilon_{\mathrm{it}} \qquad [2]$$

with:

$$\beta_{ki} = \beta_k + v_{ki}, \qquad \lambda_{ki} = \lambda_k + w_{ki}$$

Third, based on the estimated country-specific coefficients, a cluster analysis identifies systematic patterns of heterogeneity. In particular, the vectors of random coefficients $(\beta_{1i}, \beta_{2i}, \ \lambda_{1i}, \lambda_{2i})$ are used as inputs in a k-means clustering algorithm, which partitions countries into groups that minimise within-cluster variance and maximise between-cluster differences. The number of clusters is selected based on an exploratory hierarchical clustering analysis, by inspecting the corresponding dendrogram to identify meaningful breaks in the data's similarity structure. This combined approach allows us to determine an empirically grounded number of clusters while retaining the flexibility and interpretability of k-means classification.

The final step consists of re-estimating the ECM, allowing for discrete heterogeneity across clusters identified in equation [2]. This approach replaces continuous random heterogeneity with a grouped structure that improves interpretability while preserving the main patterns of cross-country differences.

The ECM with clusters is then specified as:

$$\Delta \mathrm{exp}_{\mathrm{it}} = \alpha\ \mathrm{exp}_{\mathrm{it-1}} + \textstyle\sum_c D_{ic}\, \beta_{1\mathrm{c}} \mathrm{pc65}_{\mathrm{it-1}} + \sum_c D_{ic}\, \beta_{2\mathrm{c}} \mathrm{p80}_{\mathrm{it-1}} + \sum_c D_{ic}\, \lambda_{1\mathrm{c}} \Delta \mathrm{pc65}_{\mathrm{it}} + \\ + \textstyle\sum_c D_{ic}\, \lambda_{2\mathrm{c}} \Delta \mathrm{p80}_{\mathrm{it}} + \delta' \Delta \mathrm{X}_{\mathrm{it}} + \delta \mathrm{COVID}_{\mathrm{t}} + \mu_{\mathrm{i}} + \varepsilon_{\mathrm{it}} \qquad [3]$$

Where:

- D_{ic} are cluster dummies (equal to 1 if country *i* belongs to cluster *c*).
- β_{c} are cluster-specific parameters capturing long-run effects through differenced variables (Δ)

[3] The results of those instrumental estimates are not reported in the paper.

[4] We allow slope heterogeneity only for the theoretically central variables to avoid overparameterizing the model. Given the limited cross-sectional dimension of the panel, estimating random slopes for all covariates would require many additional variance-covariance parameters and could lead to unstable estimates. This restricted mixed-effects specification provides a compromise between a fully homogeneous model and a fully heterogeneous one: it allows the key effects to vary across countries while maintaining a parsimonious structure for the remaining controls.

- λ_c are cluster-specific parameters capturing short-run effects through lagged levels $(it-1)$

Countries within the same cluster share similar adjustment patterns; differences in expenditure dynamics are captured at the group level rather than the individual level; the ECM structure ensures a consistent distinction between short-run shocks and long-run equilibrium adjustment.

4. ECONOMETRIC RESULTS

Table 2 presents the main summary statistics for all the variables. The data reveal substantial cross-country heterogeneity in both fiscal outcomes and demographic structures. Total public age-related expenditure averages around one-fifth of GDP, although the dispersion is considerable across countries and over time. Among the individual components, pensions represent the largest share, followed by healthcare, while long-term care remains comparatively small but is increasingly important in more advanced ageing contexts. Demographic indicators also exhibit notable variation. The share of the population aged 65 and over has increased steadily across all countries, while the proportion of individuals aged 80 and over within the elderly population shows even greater heterogeneity, reflecting differences in longevity and population structure. Macroeconomic variables exhibit significant variability as well, particularly in economic growth and fiscal balances, which are strongly influenced by business cycle fluctuations and major economic events, including the global financial crisis and the COVID-19 pandemic. Overall, the descriptive statistics highlight both cross-sectional and temporal variation.

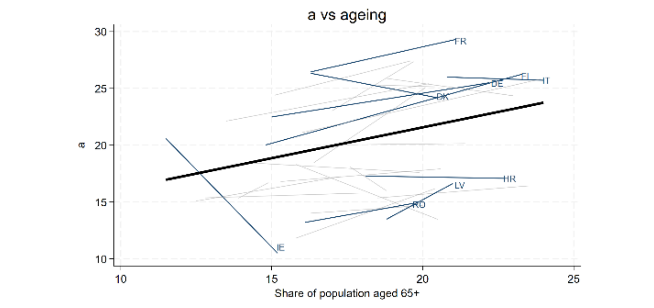
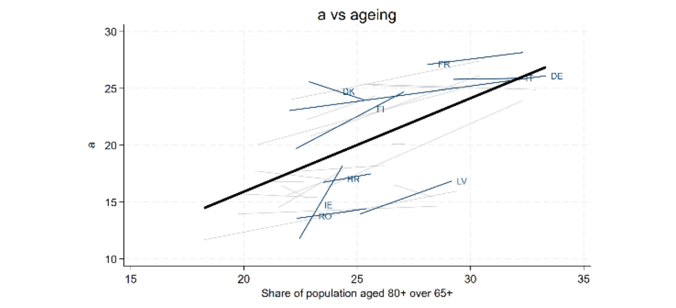
Moreover, **Figure 1** displays the estimated relationship between expenditure and the share of the population aged 65+ and 80+. The graphical evidence confirms heterogeneity across countries. While some of them show a strong positive association between ageing and expenditure, others display flatter or even weak relationships. This heterogeneity is observed in both the broader elderly population (65+) and for the oldest cohorts (80+), although the latter tends to show stronger associations in certain expenditure categories, particularly healthcare and long-term care.

Table 2. Descriptive statistics

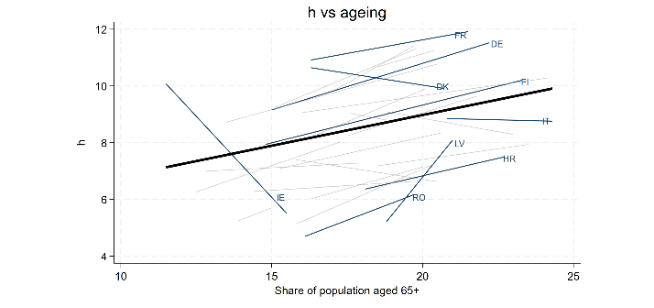
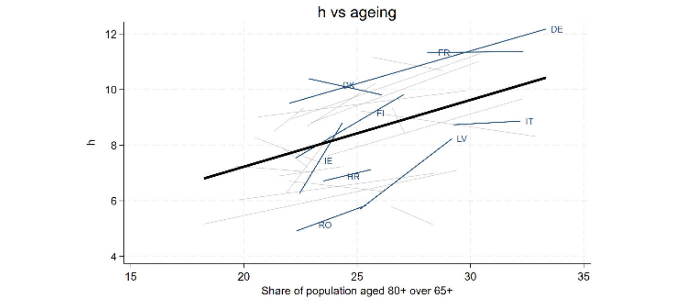
	N	Average-T	Mean	Std. Dev.	Min	Max
Total public age-related expenditure (% of GDP)	451	17	20.58	4.70	11.19	30.14
Public expenditure on Healthcare (% of GDP)	461	17	8.59	1.81	4.47	12.70
Public expenditure on Long-Term Care (% of GDP)	456	17	1.13	0.83	0.01	3.20
Public expenditure on Pensions (% of GDP)	744	28	10.37	2.80	3.66	18.17
Share of population aged 65 and over	882	33	16.56	3.09	10.30	24.30
Share of population aged 80 and over relative to population aged 65+	881	33	23.75	3.93	12.98	33.33
Real GDP growth rate (percentage point change)	793	29	2.56	3.73	-16.04	24.62
GDP per capita in PPS	810	30	24,884	13,638	4,477	97,689
General government expenditure (% of GDP)	798	30	45.47	7.05	20.70	64.90
Net lending / net borrowing (% of GDP)	808	30	-2.65	3.49	-32.10	6.90

Figure 1. Slope heterogeneity

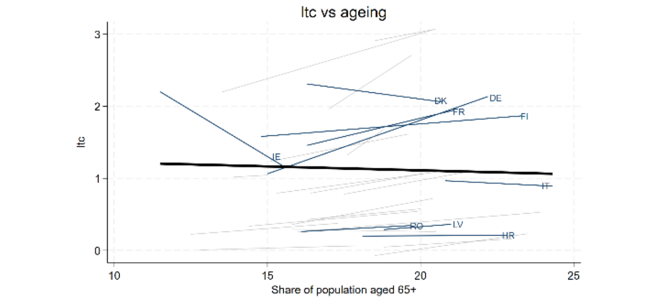
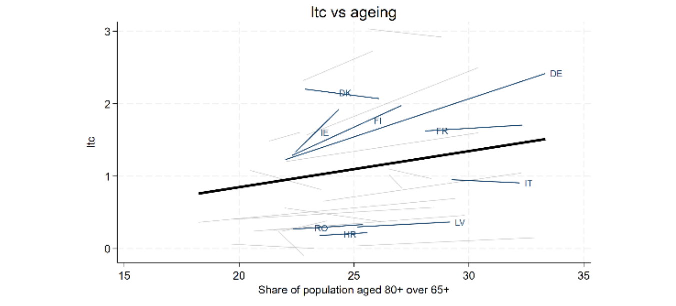
Total public age-related expenditure (% of GDP)

Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	

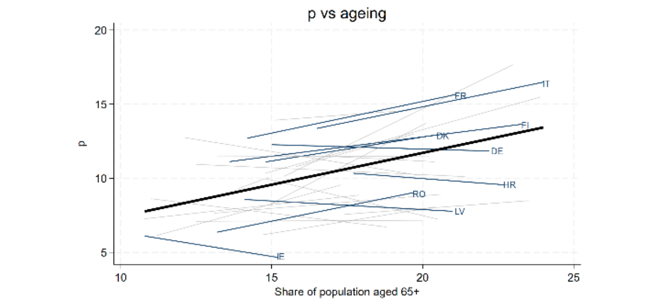
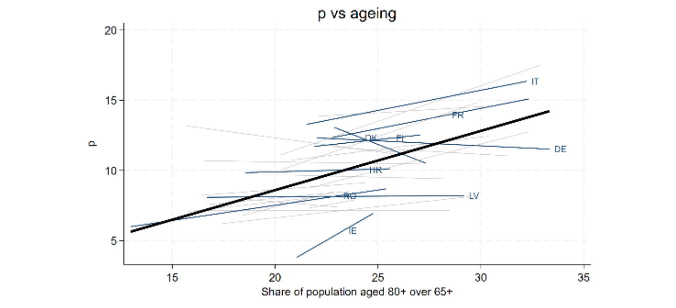
Public expenditure on Healthcare (% of GDP)

Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	

Public expenditure on Long-Term Care (% of GDP)

Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	

Public expenditure on Pensions (% of GDP)

Share of population aged 65 and over	*Share of population aged 80 and over relative to population aged 65+*
	

To test for unit roots in the left-hand variables, we rely upon a Fisher-type panel test based on augmented Dickey–Fuller regressions (**Table 3**). This approach is appropriate for our unbalanced TSCS panel because it allows for heterogeneous dynamics across countries while remaining parsimonious in a short time dimension. In all cases except for p, the test statistics lead us to reject the null of a unit root at conventional significance levels, indicating stationarity. For p, however, the evidence is weaker and compatible with a strong persistent, near-integrated process (the unit root null cannot be rejected at the 10 per cent level), which further supports the use of an error-correction specification to model both short-run dynamics and gradual adjustment toward a long-run equilibrium relationship.

Table 3. Unit Root test results

ADF-Fisher	**a**	**h**	**ltc**	**p**
p-value	0.0001	0.0000	0.0217	0.1067
Statistic	3.7107	5.5875	2.0200	1.2444

Notes: Fisher-type panel unit root tests based on ADF regressions with 1 lag. Null hypothesis: all panels contain a unit root. Modified inverse Chi-squared statistics and p-values are reported. Panels are unbalanced.

Table 4 reports results for equations [1] and [3] and the four expenditure categories. **Figure 2** shows the corresponding dendrograms for all the expenditure functions analysed, using Ward's hierarchical method based on the coefficients from the non-reported mixed-effects model in equation [2]. Finally, **Figure 3** displays the resulting EU27 maps and **Table 5** groups countries by cluster and expenditure category.

Table 4. Estimation results

	age-related expenditure [a]				Healthcare expenditure [h]				Long-term-care expenditure [ltc]				Pension expenditure [p]			
	ECM [1]		Clustered ECM [3]		ECM [1]		Clustered ECM [3]		ECM [1]		Clustered ECM [3]		ECM [1]		Clustered ECM [3]	
Lag	-0.170		-0.225		-0.253		-0.297		-0.133		-0.159		-0.124		-0.139	
	(-6.77)	***	(-8.07)	***	(-6.35)	***	(-6.45)	***	(-5.26)	***	(-6.45)	***	(-6.30)	***	(-6.18)	***
Δpc65	0.404				0.118				-0.018				0.236			
	(2.04)	*			(1.46)				(-0.70)				(2.02)	*		
Δp80	0.040				0.034				-0.005				0.023			
	(0.95)				(1.27)				(-0.71)				(0.73)			
L.pc65	0.018				0.020				0.003				-0.026			
	(0.56)				(1.15)				(1.26)				(-2.39)	**		
L.p80	0.024				0.020				0.003				0.000			
	(1.85)	*			(2.30)	**			(1.22)				(0.03)			
growth	-0.172		-0.171		-0.054		-0.054		-0.009		-0.009		-0.108		-0.109	
	(-18.73)	***	(-18.31)	***	(-7.79)	***	(-7.14)	***	(-4.16)	***	(-5.01)	***	(-16.16)	***	(-16.31)	***
L.gdp	-0.000		-0.000		-0.000		-0.000		0.000		0.000		-0.000		-0.000	
	(-1.81)	*	(-2.23)	**	(-0.66)		(-0.99)		(1.04)		(1.21)		(-0.66)		(-1.01)	
L.exp_gdp	-0.007		0.014		0.010		0.006		0.001		0.001		0.004		0.001	
	(-0.35)		(0.57)		(0.93)		(0.49)		(0.43)		(0.39)		(0.43)		(0.09)	
L.def	0.013		0.032		0.021		0.011		0.004		0.004		0.007		0.001	
	(0.71)		(1.41)		(2.31)	**	(1.13)		(1.75)	*	(1.40)		(0.67)		(0.19)	
covid	0.692		0.727		0.560		0.543		0.016		0.014		0.115		0.123	
	(7.74)	***	(8.03)	***	(8.07)	***	(7.37)	***	(1.42)		(1.24)		(2.53)	**	(2.90)	***
Δpc65 by clusters																
1			0.775				0.162				-0.003				0.131	
			(2.76)	**			(1.27)				(-0.11)				(2.10)	**
2			0.701				0.022				0.043				0.485	
			(1.69)				(0.11)				(0.66)				(4.18)	***
3			0.076				0.155				-0.078				0.628	
			(0.41)				(0.71)				(-1.66)				(5.12)	***
4			-1.090				0.058				-0.010				-0.487	

	age-related expenditure [a]		Healthcare expenditure [h]		Long-term-care expenditure [ltc]		Pension expenditure [p]	
	ECM [1]	Clustered ECM [3]	ECM [1]	Clustered ECM [3]	ECM [1]	Clustered ECM [3]	ECM [1]	Clustered ECM [3]
		(-3.90) ***		(0.29)		(-0.20)		(-3.19) ***
Δp80 by clusters								
1		0.061		0.077		-0.011		-0.049
		(1.50)		(3.24) ***		(-2.21) **		(-2.97) ***
2		-0.165		-0.116		-0.045		0.115
		(-0.96)		(-4.38) ***		(-4.00) ***		(3.58) ***
3		0.112		0.154		-0.113		-0.137
		(1.19)		(2.71) **		(-1.35)		(-2.57) **
4		0.223		-0.071		0.028		0.088
		(1.73) *		(-3.90) ***		(7.19) ***		(1.18)
L.pc65 by clusters								
1		0.064		0.038		0.004		-0.031
		(1.82) *		(2.02) *		(0.90)		(-2.97) ***
2		-0.067		0.009		-0.020		-0.018
		(-0.90)		(0.53)		(-1.13)		(-1.11)
3		0.005		0.070		-0.013		-0.015
		(0.13)		(2.71) **		(-0.76)		(-0.99)
4		-0.001		-0.023		0.005		0.000
		(-0.02)		(-0.71)		(1.90) *		(0.01)
L.p80 by clusters								
1		0.065		0.046		0.002		0.014
		(1.79) *		(3.65) ***		(0.75)		(0.93)
2		0.038		0.015		0.007		0.005
		(1.24)		(0.95)		(1.07)		(0.43)
3		0.035		0.016		-0.040		-0.005
		(2.12) **		(1.19)		(-0.51)		(-0.66)
4		-0.022		0.031		0.007		-0.005
		(-0.26)		(2.46) **		(1.41)		(-0.14)
_cons	3.639	3.347	1.081	1.266	-0.014	0.056	1.836	1.914
	(4.52) ***	(3.25) ***	(2.21) **	(2.46) **	(-0.11)	(0.27)	(5.48) ***	(5.80) ***
N	421	421	431	431	426	426	686	686
Countries	27	27	27	27	27	27	27	27
Average T	15.59	15.59	15.96	15.96	15.78	15.78	25.41	25.41

	age-related expenditure [a]		Healthcare expenditure [h]		Long-term-care expenditure [ltc]		Pension expenditure [p]	
	ECM [1]	**Clustered ECM [3]**	**ECM [1]**	**Clustered ECM [3]**	**ECM [1]**	**Clustered ECM [3]**	**ECM [1]**	**Clustered ECM [3]**
Max VIF	3.49	3.49	2.21	2.21	1.91	1.91	2.77	2.77
R^2 within	0.75	0.77	0.64	0.66	0.30	0.36	0.64	0.67
R^2 overall	0.04	0.13	0.05	0.17	0.02	0.00	0.00	0.01
R^2 between	0.36	0.23	0.27	0.16	0.02	0.01	0.47	0.49

Notes: t-statistics in parentheses. *** p<.01, ** p<.05, * p<.1. Max VIF excludes the lagged dependent variabl

Figure 2. Dendrogram based on mixed-effects ECM coefficients

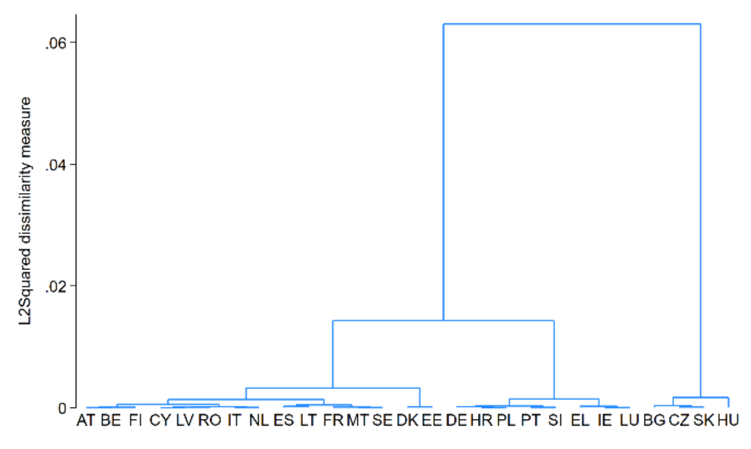
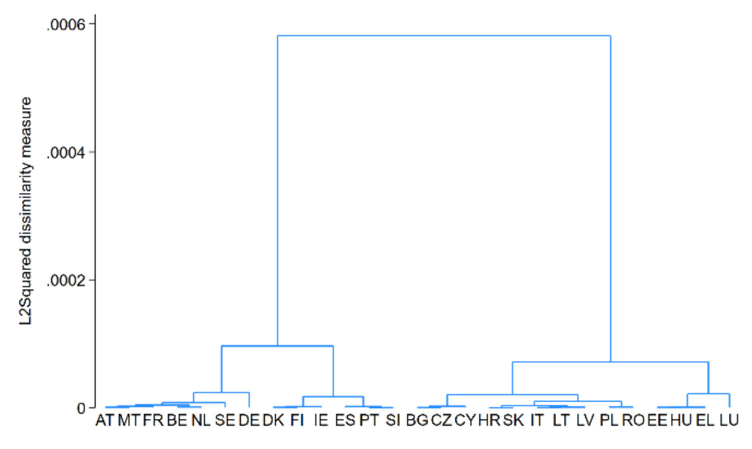
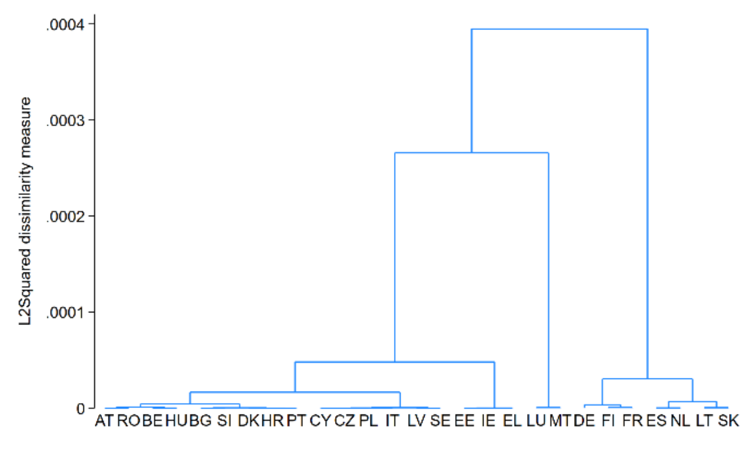
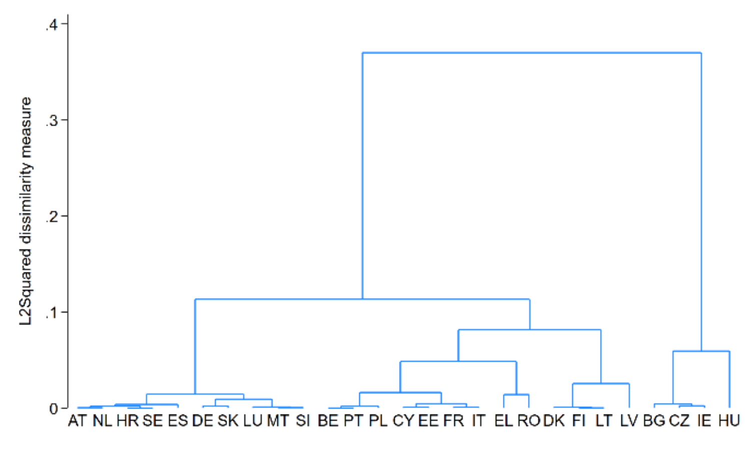
Total public age-related expenditure (% of GDP)	**Public expenditure on Healthcare (% of GDP)**
	
Public expenditure on Long-Term Care (% of GDP)	**Public expenditure on Pensions (% of GDP)**
	

Figure 3. Maps of clustered countries

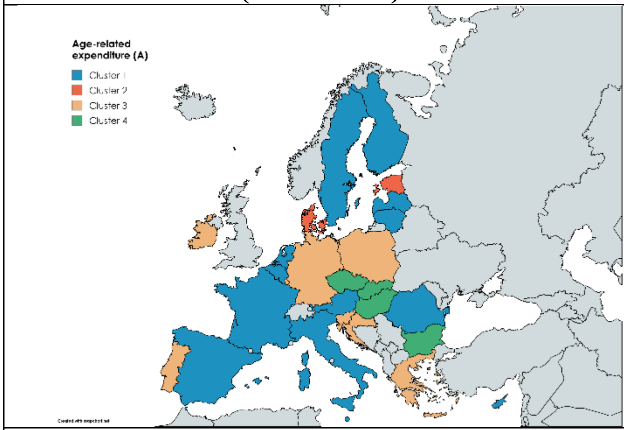
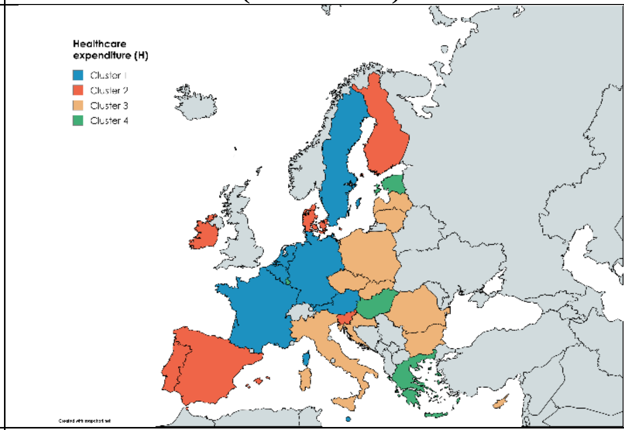
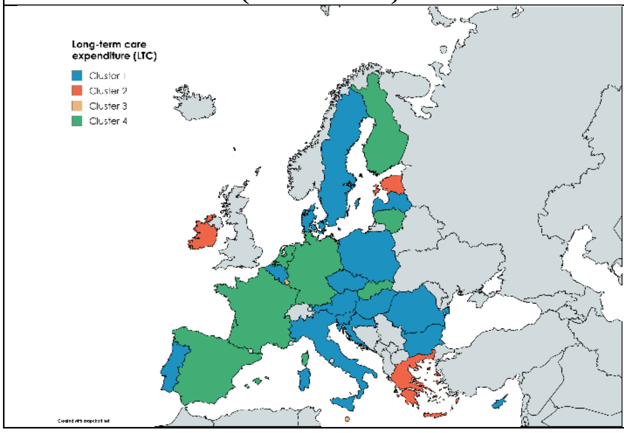
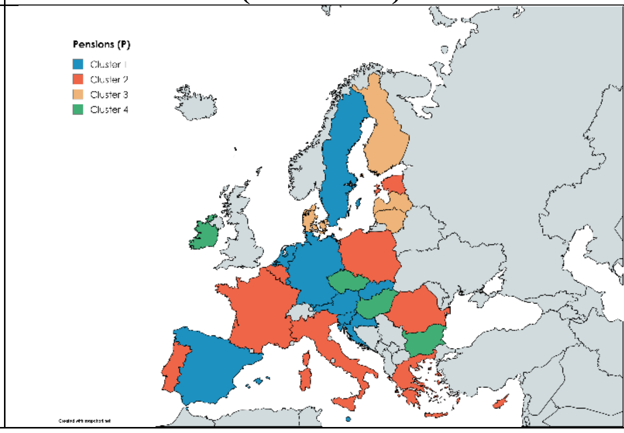
Total public age-related expenditure (% of GDP)	Public expenditure on Healthcare (% of GDP)
	
Public expenditure on Long-Term Care (% of GDP)	**Public expenditure on Pensions (% of GDP)**
	

Table 5. Country clusters by expenditure category

	Aggregate (A)	Healthcare (H)	Long-Term-Care (LTC)	Pensions (P)
Cluster 1	Austria (AT), Belgium (BE), Cyprus (CY), Spain (ES), Finland (FI), France (FR), Italy (IT), Lithuania (LT), Latvia (LV), Malta (MT), Netherlands (NL), Romania (RO), Sweden (SE)	Austria (AT), Belgium (BE), Germany (DE), France (FR), Malta (MT), Netherlands (NL), Sweden (SE)	Austria (AT), Belgium (BE), Bulgaria (BG), Cyprus (CY), Czechia (CZ), Denmark (DK), Croatia (HR), Hungary (HU), Italy (IT), Latvia (LV), Poland (PL), Portugal (PT), Romania (RO), Sweden (SE), Slovenia (SI)	Austria (AT), Germany (DE), Spain (ES), Croatia (HR), Luxembourg (LU), Malta (MT), Netherlands (NL), Sweden (SE), Slovenia (SI), Slovakia (SK)
Cluster 2	Denmark (DK), Estonia (EE)	Denmark (DK), Spain (ES), Finland (FI), Ireland (IE), Portugal (PT), Slovenia (SI)	Estonia (EE), Greece (EL), Ireland (IE)	Belgium (BE), Cyprus (CY), Estonia (EE), Greece (EL), France (FR), Italy (IT), Poland (PL), Portugal (PT), Romania (RO)
Cluster 3	Germany (DE), Greece (EL), Croatia (HR), Ireland (IE), Luxembourg (LU), Poland (PL), Portugal (PT), Slovenia (SI)	Bulgaria (BG), Cyprus (CY), Czechia (CZ), Croatia (HR), Italy (IT), Lithuania (LT), Latvia (LV), Poland (PL), Romania (RO), Slovakia (SK)	Luxembourg (LU), Malta (MT)	Denmark (DK), Finland (FI), Lithuania (LT), Latvia (LV)
Cluster 4	Bulgaria (BG), Czechia (CZ), Hungary (HU), Slovakia (SK)	Estonia (EE), Greece (EL), Hungary (HU), Luxembourg (LU)	Germany (DE), Spain (ES), Finland (FI), France (FR), Lithuania (LT), Netherlands (NL), Slovakia (SK)	Bulgaria (BG), Czechia (CZ), Hungary (HU), Ireland (IE)

Note: Clusters are derived from hierarchical clustering (Ward's method) applied to country-specific coefficients extracted from the mixed-effects ECM model.

Age-Related Expenditure aggregate (a)

The baseline ECM results confirm that public expenditure displays a high degree of inertia, as reflected in the negative and highly significant coefficient on the lagged dependent variable (−0.170). In the short run, increases in the share of the population aged 65 ($\Delta pc65_{it}$) exert a positive effect, while the impact of the 80+ population is not statistically significant. Long-run effects are weaker, with only the 80+ variable showing significance (p-value<0.10). Overall, these findings suggest that demographic pressures operate gradually, with limited immediate effects and a strong role for fiscal inertia. Looking at differences across countries, four clusters of countries emerge:

- **Cluster 1:** Austria (AT); Belgium (BE); Finland (FI); Cyprus (CY); Latvia (LV); Romania (RO); Italy (IT); Netherlands (NL); Spain (ES); Lithuania (LT); France (FR); Malta (MT); Sweden (SE). The clustered ECM results show a strong and statistically significant short-run effect of ageing (65+), indicating that increases in the elderly population are rapidly translated into higher expenditure. The short-run effect of advanced ageing (80+) is not statistically significant, suggesting that the composition of ageing does not play a relevant role in immediate expenditure dynamics. In the long run, both demographic variables also display positive coefficients but are only weakly significant (p-values < 0.10). Hence, this cluster, which includes the largest number of countries, would follow the patterns detected for the whole panel.

- **Cluster 2**: Denmark (DK) and Estonia (EE). Neither demographic variable is statistically significant in either the short or long run, indicating that ageing, regardless of its intensity, has no consistent or stable association with expenditure.
- **Cluster 3:** Germany (DE); Croatia (HR); Poland (PL); Portugal (PT); Slovenia (SI); Greece (EL); Ireland (IE); Luxembourg (LU). The clustered ECM reveals no short-run effects for either demographic variable. In contrast, the 80+ population shows a positive and statistically significant long-run effect, suggesting that advanced ageing gradually shapes expenditure dynamics. No such long-run effect is found for the broader 65+ population.
- **Cluster 4:** Bulgaria (BG); Czechia (CZ); Slovakia (SK); Hungary (HU). The clustered ECM results reveal a distinctive and asymmetric adjustment pattern. In the short run, a higher share of the elderly population is associated with significantly lower expenditure, suggesting that demographic pressure may induce fiscal restraint. The effect of advanced ageing is positive but only weakly significant. In the long run, neither variable is statistically significant.

Healthcare Expenditure (h)

Short-run effects of demographic variables are not statistically significant in the baseline ECM. In contrast, long-run effects for the 80+ population are positive and statistically significant. Macroeconomic variables behave as expected, with growth exerting a negative effect and the COVID-19 dummy capturing a strong and significant increase in spending. Four groups of countries are detected:

- **Cluster 1**: Austria (AT); Malta (MT); France (FR); Belgium (BE); Netherlands (NL); Sweden (SE); Germany (DE). The clustered ECM results reveal a clear and consistent role for advanced ageing. In the short run, the effect of the 65+ population is not statistically significant, whereas the coefficient on $\Delta p80_{it}$ is positive and highly significant. Again, the coefficient on the 65+ population is smaller and only weakly significant, but the lagged 80+ population shows a positive and statistically significant impact.
- **Cluster 2**: Denmark (DK); Finland (FI); Ireland (IE); Spain (ES); Portugal (PT); Slovenia (SI). The clustered ECM results reveal a highly selective and asymmetric response to demographic change. In the short run, the coefficient on $\Delta pc65_{it}$ is not statistically significant, the effect of $\Delta p80_{it}$ is negative and highly significant. In the long run, no statistically significant relationship is observed for either demographic variable.
- **Cluster 3:** Bulgaria (BG); Czechia (CZ); Cyprus (CY); Croatia (HR); Slovakia (SK); Italy (IT); Lithuania (LT); Latvia (LV); Poland (PL); Romania (RO). Results show a differentiated response to demographic change across time horizons. In the short run, the coefficient on $\Delta pc65_{it}$ is not statistically significant, while the effect of advanced ageing ($\Delta p80_{it}$) is positive and significant. In contrast, the long-run adjustment follows a different pattern. The coefficient on the lagged 65+ population is positive and statistically significant, whereas the coefficient on the 80+ variable is not.

- **Cluster 4:** Estonia (EE); Hungary (HU); Greece (EL); Luxembourg (LU). The clustered ECM results reveal a clear but asymmetric response to demographic change. In the short run, the coefficient on $\Delta p80_{it}$ is negative and statistically significant, but it changes to positive in the long run. By contrast, no significant effect is observed for the broader 65+ population in either the short or long run.

Unexpected negative coefficients for Cluster 2 and Cluster 4 could reflect different mechanisms. For Cluster 2, the negative short-run effect on the 80+ population suggests that countries with more developed public healthcare systems have greater capacity to implement efficiency measures as demographic pressure increases, preventing public expenditure from rising in proportion. In Cluster 4, the negative-to-positive sign would reflect a delayed adjustment: short-run fiscal constraints temporarily prevent the expenditure response to demographic pressure, but long-run equilibrium converges to the expected positive relationship as institutional adjustments overcome financing constraints.

Long-Term Care Expenditure (ltc)

In the baseline ECM, demographic effects are not statistically significant in either the short or long run. Things change slightly when the analysis is broken down into country groups.

- **Cluster 1**: Austria (AT); Romania (RO); Belgium (BE); Hungary (HU); Bulgaria (BG); Slovenia (SI); Denmark (DK); Croatia (HR); Portugal (PT); Cyprus (CY); Czechia (CZ); Poland (PL); Italy (IT); Latvia (LV); Sweden (SE). In the short run, the coefficient on $\Delta p80_{it}$ is negative and statistically significant, indicating that increases in the oldest population are associated with immediate expenditure restraint. In the long run, neither demographic variable shows a statistically significant effect, suggesting no stable structural link between ageing and LTC expenditure.
- **Cluster 2**: Estonia (EE); Ireland (IE); Greece (EL). In the short run, the coefficient on $\Delta p80_{it}$ is again negative and highly significant, but much larger than in the case of Cluster 1. In the long run, neither demographic variable has a statistically significant effect.
- **Cluster 3:** Luxembourg (LU); Malta (MT). In the short run, the coefficient on the 65+ population is negative and marginally significant. The short-run effect of advanced ageing ($\Delta p80_{it}$) is also negative but statistically insignificant. In the long run, neither demographic variable displays a statistically significant coefficient.
- **Cluster 4:** Germany (DE); Finland (FI); France (FR); Spain (ES); Netherlands (NL); Lithuania (LT); Slovakia (SK). In the short run, the coefficient on $\Delta p80_{it}$ is positive and highly significant. In the long run, evidence of demographic effects is weaker. The coefficient on the lagged 65+ population is positive but only marginally significant, while the 80+ variable is not statistically significant.

Pension Expenditure (p)

Results for the baseline ECM indicate limited short-run effects, with only the 65+ population showing a weak positive effect (p-value < 0.10). In the long run, the coefficient on 65+ is negative and statistically significant. Results for clusters are the following:

- **Cluster 1:** Austria (AT); Netherlands (NL); Croatia (HR); Sweden (SE); Spain (ES); Germany (DE); Slovakia (SK); Luxembourg (LU); Malta (MT); Slovenia (SI). In the short run, the coefficient on $\Delta pc65_{it}$ is positive and statistically significant. In contrast, the effect of advanced ageing ($\Delta p80_{it}$) is negative and significant. In the long run, the dominant result is a negative and highly significant coefficient on the lagged 65+ population. This indicates the presence of structural containment mechanisms, whereby pension systems adjust over time to offset the fiscal impact of population ageing. No significant long-run effect is observed for the 80+ population.
- **Cluster 2:** Belgium (BE); Portugal (PT); Poland (PL); Cyprus (CY); Estonia (EE); France (FR); Italy (IT); Greece (EL); Romania (RO). T. Both $\Delta pc65_{it}$ and $\Delta p80_{it}$ exhibit positive and statistically significant coefficients, indicating that increases in both general and advanced ageing are immediately translated into higher pension expenditure. This suggests a high degree of short-term demographic responsiveness, with limited buffering through institutional mechanisms. In contrast, no statistically significant long-run effects are observed.
- **Cluster 3:** Denmark (DK); Finland (FI); Lithuania (LT); Latvia (LV). The coefficient on $\Delta pc65_{it}$ is large and highly significant, indicating that increases in the elderly population are rapidly and substantially reflected in pension expenditure. At the same time, the coefficient on $\Delta p80_{it}$ is negative and statistically significant, pointing to offsetting pressures between different segments of the ageing population. In contrast, no significant long-run effects are observed for either demographic variable, indicating the absence of a structural adjustment mechanism linking ageing to expenditure over time.
- **Cluster 4:** Bulgaria (BG); Czechia (CZ); Ireland (IE); Hungary (HU). The clustered ECM results reveal a distinct and asymmetric adjustment pattern. In the short run, the coefficient on $\Delta pc65_{it}$ is negative and highly significant, indicating that increases in the elderly population are associated with immediate expenditure restraint rather than expansion. By contrast, the short-run effect of advanced ageing ($\Delta p80_{it}$) is not statistically significant, suggesting no clear role for the oldest population in short-term pension dynamics. In the long run, neither demographic variable displays a statistically significant effect, indicating the absence of a stable structural relationship between ageing and pension expenditure.

The joint results for healthcare, long-term care (LTC), and pension expenditure reveal three main patterns. First, demographic effects are markedly heterogeneous across countries and are largely driven by short-run responses, although their intensity varies by spending category, being more limited in healthcare and virtually absent in LTC, underscoring the rigid or structural nature of these systems. Second, advanced ageing (80+) emerges as the key driver of long-run dynamics, although its impact is uneven and more pronounced in countries with more developed welfare systems, particularly in LTC and, to a lesser extent, in healthcare, while its relevance in pensions is more diffuse.

Third, the cluster analysis points to a certain geographical and institutional structure: Central and Eastern European economies tend to exhibit weaker or more delayed responses,[5] whereas Western and Nordic countries tend to show stronger and more persistent adjustments.[6] This pattern is more evident in healthcare and LTC than in pensions, where expenditure shows greater persistence and heterogeneity in adjustment patterns, reflecting the decisive role of institutional design in shaping public spending responses to ageing.

5. CONCLUSIONS

A central result of this paper is that the fiscal impact of demographic ageing is highly heterogeneous across expenditure categories, demographic dimensions, time horizons, and countries. Rather than following a uniform or mechanical pattern, the results reveal a set of distinct and sometimes opposing adjustment mechanisms, reflecting differences in institutional design, policy frameworks, and fiscal capacity.

A first robust finding across all specifications is the strong persistence of public expenditure. This persistence is particularly pronounced in pensions and healthcare, where expenditure is shaped by legal commitments and structural entitlements. By contrast, long-term care exhibits weaker, less stable dynamics, consistent with a more heterogeneous and evolving institutional structure.

A second key result concerns the differentiated role of demographic variables. The 65+ population is primarily associated with short-run dynamics, especially in pensions. In contrast, the 80+ population captures advanced ageing and care intensity. It is a key driver in healthcare, often affecting both short- and long-run dynamics, while in long-term care and pensions, its impact is more fragmented and frequently limited to the short run. By comparison, its effects on long-term care and pension expenditure are considerably weaker, less systematic, and highly dependent on country -specific institutional arrangements. These findings suggest that the composition of ageing matters, although its fiscal implications vary substantially across expenditure categories.

Third, the clustering analysis identifies several differentiated expenditure-response patterns, ranging from expansionary and reactive responses to more constrained or weakly responsive trajectories. These results suggest that demographic pressures are filtered through heterogeneous institutional and fiscal contexts rather than producing uniform expenditure outcomes. Despite this heterogeneity, some regularities emerge. Several Western and Northern European countries consistently cluster together across healthcare, long-term care, and pension expenditure, reflecting broadly similar welfare-state structures and stronger institutional capacity to respond to demographic pressures. In contrast, many Central and Eastern European countries tend to form a distinct group characterised by weaker or more delayed expenditure responses, consistent with more limited fiscal capacity and less consolidated welfare arrangements. However, this structural stability is only partial. Cluster membership varies substantially across expenditure categories, particularly in long-term care and pensions, where institutional designs differ more markedly across countries. Smaller states, transition economies, and

[5] Bulgaria (BG), Czechia (CZ), Hungary (HU), Slovakia (SK), and, in an outer, less cohesive circle: Latvia (LV), Lithuania (LT), Poland (PL), Croatia (HR), and Romania (RO)
[6] Austria (AT), Belgium (BE), Denmark (DK), Finland (FI), and, in a second, more permissive circle, France (FR), Germany (DE), the Netherlands (NL), and Sweden (SE).

countries with hybrid welfare configurations exhibit the highest cluster mobility, frequently shifting across groups depending on the expenditure category considered.

Our results challenge the view of ageing as a uniform fiscal burden. Instead, they show that demographic change operates as a conditional and institutionally mediated driver whose impact depends on the capacity of public systems to absorb, adjust to, or offset demographic pressures. In this sense, fiscal sustainability under ageing emerges as a fundamentally institutional outcome, rather than a purely demographic one.

Understanding the divergent patterns presented in this paper requires moving beyond aggregate analyses and working with microdata employing more sophisticated econometric techniques, such as regression discontinuity designs. It would be particularly valuable for identifying whether the practice of using universal cut-off points, such as ages 65 and 80, is meaningful or whether they vary systematically across country groups and expenditure functions. Moreover, future research should delve into identifying the specific policy mechanisms, such as pension indexation rules, healthcare financing models, or the degree of formalization of long-term care, that reinforce these different adjustment regimes and explain the observed cross-country heterogeneity. However, a more fundamental step is to recognize and systematically analyze the diversity across country groups shown in this paper.

REFERENCES

Acemoglu, D., & Restrepo, P. (2022). "Demographics and automation". *The Review of Economic Studies*, 89 (1):1-44. DOI: https://doi.org/10.1093/restud/rdab031

Aguilar-Segado, C.D (2024). "El Régimen Financiero y Tributario de la Pensión Unificada en la Unión Europea", *Revista Jurídica de Investigación e Innovación Educativa*, 129-150. DOI: https://doi.org/10.24310/rejie.30.2024.18344

Ajovalasit, S., Consiglio, A. and Provenzano, C. (2024). "Debt Sustainability in the Context of Population Ageing: A Risk Management Approach", *Risks,* 12(188). DOI: https://doi.org/10.3390/risks12120188

Aksoy, Y., Basso, H., Smith, R.., Grasl, T. (2019). "Demographic Structure and Macroeconomic Trends". *American Economic Journal: Macroeconomics*, 11(1): 193-222. DOI: 10.1257/mac.20170114.

Auerbach, A. J., Gokhale, J., & Kotlikoff, L.J. (1994). "Generational Accounting: A Meaningful Way to Evaluate Fiscal Policy." *Journal of Economic Perspectives*, 8(1): 73–94. DOI: 10.1257/jep.8.1.73.

Auerbach, A., Kotlikoff, L.J. & Leibfritz, W. (1999). "Generational Accounting Around the World" NBER Chapters in Generational Accounting around the Worl, p. 1-8, National Bureau of Economic Research, Inc.

Baltagi, B. H. (2021). *Econometric Analysis of Panel Data* (6th ed.). Wiley, Springer Texts.

Barilari, F., Mastrorocco, N., Paradisi, M. (2024). "Population Ageing, Voting and Political Agendas" SSRN. DOI. https://dx.doi.org/10.2139/ssrn.4880469

Beck, Nathaniel, and Jonathan N. Katz. 2011. "Modelling Dynamics in Time-Series–Cross-Section Political Economy Data." *Annual Review of Political Science* 14: 331–352. DOI: https://doi.org/10.1146/annurev-polisci-071510-103222

Beetsma, R., Busse, M., Larch, M., and Romp, W.E. (2025). "Some Intergenerational Arithmetic to Control Public Debt in the EU", *Journal of Pension and Economics & Finance*, 1-18. DOI: 10.1017/S1474747225100085

Benzarti, Y., & Carloni, D. (2019). "Who Really Benefits from Consumption Tax Cuts? Evidence from a Large VAT Reform in France". *American Economic Journal: Economic Policy*, 11(1): 38–63. DOI: 10.1257/pol.20170504

Blundell, R., & Bond, S. (1998). "Initial conditions and moment restrictions in dynamic panel data models". *Journal of Econometrics*, 87(1), 115–143. DOI: https://doi.org/10.1016/S0304-4076(98)00009-8

Bond, S. (2002)."Dynamic panel data models: A guide to microdata methods and practice". *Portuguese Economic Journal*, 1(2), 141–162. DOI: https://doi.org/10.1920/wp.cem.2002.0902

Brosig, M. (2022). "The Great Recession and Pension Policy Change in European Countries" *International Impacts on Social Policy*, pp. 385-398, DOI: https://doi.org/10.1007/978-3-030-86645-7_30

Brys, B., Perret, S., Thomas, A., & O'Reilly, P. (2016). "Tax Design for Inclusive Economic Growth". *OECD Taxation Working Papers* No. 26. OECD Publishing, Paris, https://doi.org/10.1787/5jlv74ggk0g7-en

Cylus, J., Williams, G., Carrino, L, Toubal, T. Barber, S. (2022). "Population ageing and health financing: A method for Forecasting two sides of the same coin", *Health Policy*, 126(12): 1226-1232. DOI: https://doi.org/10.1016/j.healthpol.2022.10.004

Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276. DOI: https://doi.org/10.2307/1913236

European Commission (2024). *Ageing Report: Underlying Assumptions and Projection Methodologies. Institutional Paper 217*. Directorate-General for Economic and Financial Affairs. DOI: 10.2765/960576

Filipović, S., and Miljković, M. (2024). "Impact of Population Ageing on Fiscal Balance in the European Union" *Stanovnistvo*, 62 (2), Institute of Social Sciences. DOI: https://doi.org/10.59954/stnv.648

IMF (2023). *Fiscal Monitor: On the Path to Policy Normalization*. International Monetary Fund, Washington D.C.

Iuga, C., and Nerisanu, and Iuga, H. (2024). "The impact of healthcare system quality and economic factors on the older adult population: a health economics perspective" *Frontiers in Public Health*, 12: 01-13. DOI: 10.3389/fpubh.2024.1454699

Michalski, T., and Stępień, J. (2021). "Ageing in European post-communist countries- is it a threat to the welfare system?" *Environmental & Socio-economic Studies*, 9 (2): 63-71. DOI: 10.2478/environ-2021-0011

Navarro-García, C. and Sarria-Santamera, A. (2023). "The Effect of Older Population on Public Health Spending: Evidence from Spain" *Health Care Science* 2(5): 306-316. DOI: 10.1002/hcs2.68

Nickell, Stephen. 1981. "Biases in Dynamic Models with Fixed Effects." *Econometrica* 49 (6): 1417–1426. DOI: https://doi.org/10.2307/1911408

OECD. (2020). *International Migration Outlook 2020*. OECD Publishing. DOI: https://doi.org/10.1787/9789264290303-en

OECD. (2021). *The Role and Design of Net Wealth Taxes in the OECD*. OECD Tax Policy Studies WP. DOI: https://doi.org/10.1787/9789264290303-en

OECD. (2022a). *Funding the Future. The Impact of Population Ageing on Revenues across Levels of Government*, OECD WP on Fiscal Federalism. DOI: https://doi.org/10.1787/22265848

OECD. (2022b). *Restoring Fiscal Sustainability in the Euro Area. Raise Taxes or Curb Spending?* OECD Economic Policy Paper 31. DOI: https://doi.org/10.1787/414711615127

OECD. (2024). Health at a Glance: Europe 2024, DOI: https://doi.org/10.1787/b3704e14-en

Petrucci, L. and Margarito, A. (2025). "Population Ageing in Italy: Demographic Trends, Economic and Healthcare Impacts, and Challenges for Welfare Sustainability" *Journal of Advanced Health Care*, 7 (4). DOI: https://doi.org/10.36017/jahc202574478

Piketty, T., & Saez, E. (2013). "Optimal Labor Income Taxation". In A. J. Auerbach, R. Chetty, M. Feldstein & E. Saez (Eds.), *Handbook of Public Economics*, 5: 391–474. Elsevier. DOI: https://doi.org/10.1016/B978-0-444-53759-1.00007-8

Ramos-Herrera, M.C. and Sosvilla-Rivero, S. (2020). “Fiscal Sustainability in Ageing Societies: Evidence from Euro Area Countries”, *Sustainability*, 12 (24), 10276. DOI: https://www.mdpi.com/2071-1050/12/24/10276

Szwarc, K. (2024). “Starzenie się społeczeństw w krajach Unii Europejskiej” *Droga do zrównoważonej gospodarki światowej* DOI: .10.18559/978-83-8211-245-0/9

Tibshirani, R. (1996). Regression shrinkage and selection via the Lasso. Journal of the Royal Statistical Society: Series B, 58(1), 267–288. DOI: https://doi.org/10.1111/j.2517-6161.1996.tb02080.x

Twarowska, M., and Twarowska-Mól, K. (2024). “Factor Influencing the Public Finance Sector Budget Balance in European Union Member States in 2004-2020” *Studia Prawno-Ekonomiczne*, CXXXI, DOI: https://doi.org/10.26485/SPE/2024/131/6

Verbič, M., Spruk, R. (2024). “Aging Population and Public Pensions: Theory and Macroeconometric Evidence” *Panoeconomicus*, 61 (3): 289-316. DOI: 10.1016/j.jpolmod.2019.07.009